PREDICTIVE MODELS IN ICU DATABASES

Alain, Darlene, Mariah
DATA 4382: DATA CAPSTONE 2

The University of Texas at Arlington
College of Science - Division of Data Science

DEPLOYMENT PROTOTYPE

ICU Risk Alert System				
Patient ID	Bad Outcome Risk %	Risk Contributors	Recommended Course of Action	Actual Outcome
12345	3.4	Local SHAP Values	Standard monitoring	0
6789	87.42	Local SHAP Values	Immediate ICU Attention	1

MODEL DEPLOYMENT RATIONALE

Strong discrimination
High ROC-AUC

Maintains real-world ICU prevalence
Class_weight="balanced" to improve learning of rare events, optional

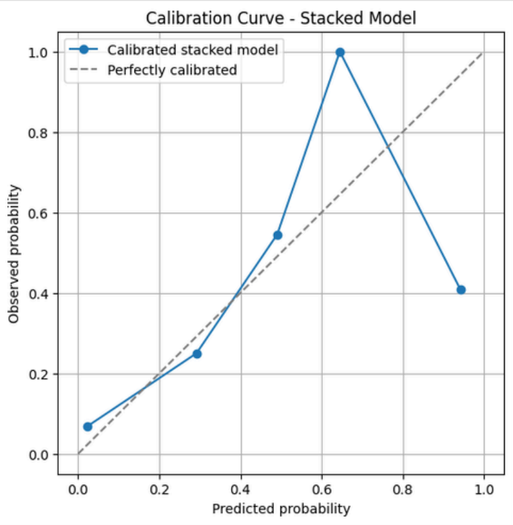
Probability calibration applied
Ensures predicted risk % reflects real ICU outcome rates

Threshold chosen for clinical goals
Balanced recall (catch high-risk patients)
Acceptable precision (avoid excessive false alarms)

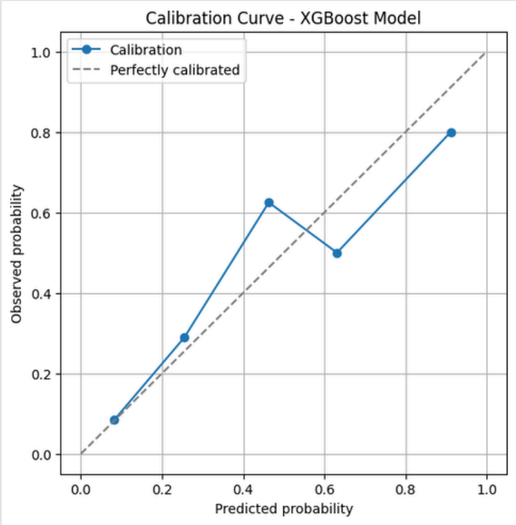
Explainability integrated
Transparent "Key Risk Drivers" provided per patient
Supports clinical decision-making

COMPARITIVE MODEL EVALUATION

Model	Class Weight	Calibrated Test ROC	Calibrated Test PRC	Calibrated Brier Score	Recall	Precision	Specificity
Stacked	Balanced	0.814	0.40	0.0943	0.627	0.356	0.849
XGBoost	Unbalanced	0.775	0.379	0.0897	0.797	0.229	0.645

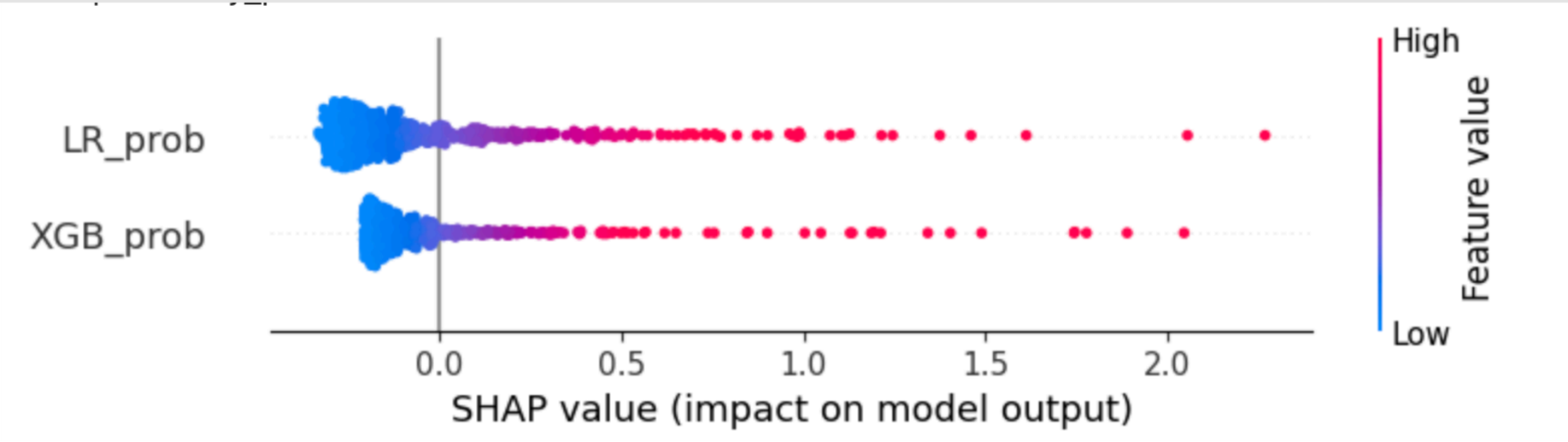


STACKED

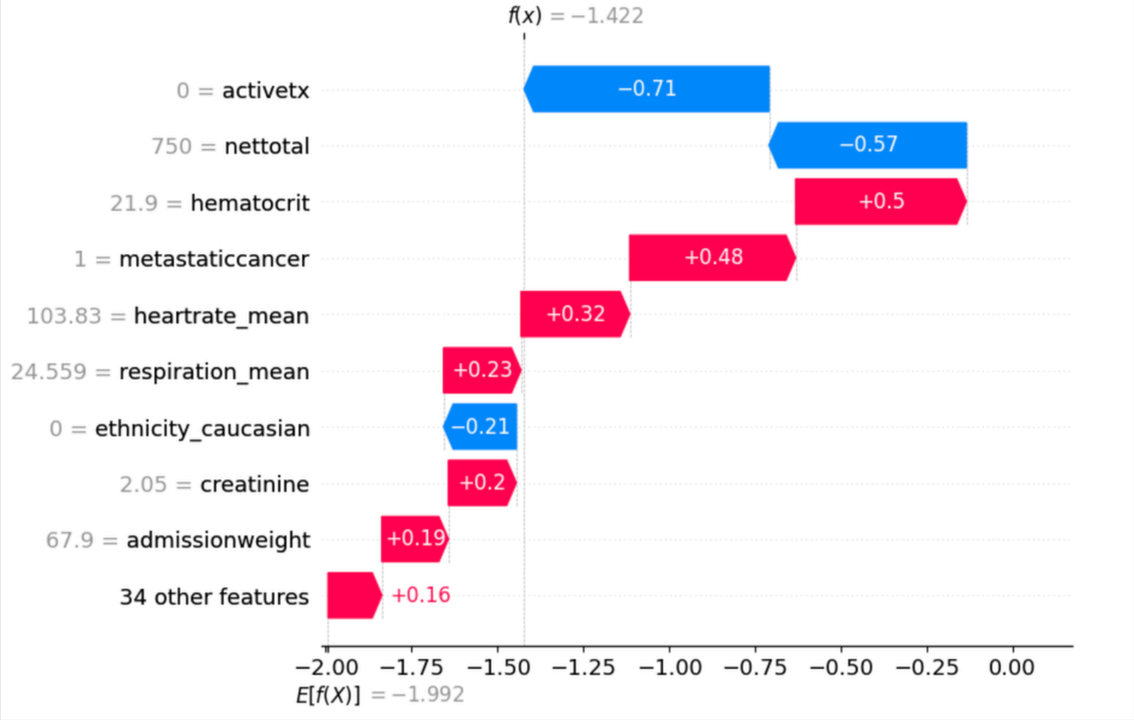


XGBOOST

GLOBAL SHAP ON STACKED



TEST PATIENT SHAP



XAI-FIRST ICU RISK INTERFACE PROTOTYPE

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Example patient index: 1547
True label: 1
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	age	admissionheight	admissionweight	apachescore	intubated	dialysis	temperature	respiratoryrate	meanbp
1547	71.0	175.3	58.0	60.0	0.0	0.0	36.2	31.0	69.0

```
This patient is LOW risk (stacked p=0.005). Top drivers from XGBoost were lower heartrate mean, higher activetx, and lower admissionweight.
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Calibrated stacked risk: 0.5%  →  LOW band
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Patient risk band: LOW
In the TEST set, patients in the LOW band had 3.8% observed mortality (n=341).
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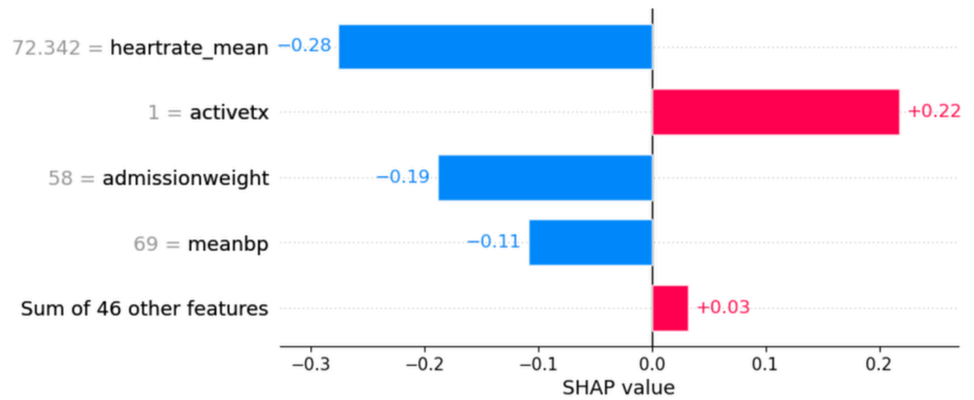
	band	n	observed_mortality
0	high	64	0.453125
1	low	341	0.038123
2	medium	99	0.171717



Interface Roadmap (Tentative)

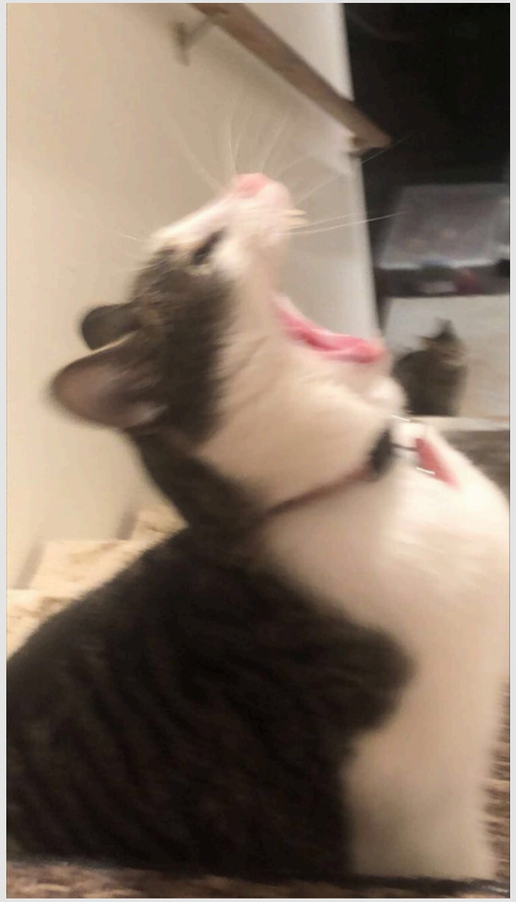
- Deploy as a clinician-facing dashboard (Streamlit-style prototype)
- Include stacked risk + risk band + SHAP explanation
- Add comparison toggle (Stacked vs XGBoost)
- Add counterfactual “Model Second Opinion” widget
- Explore partial dependence mini-plots (feature sensitivity)

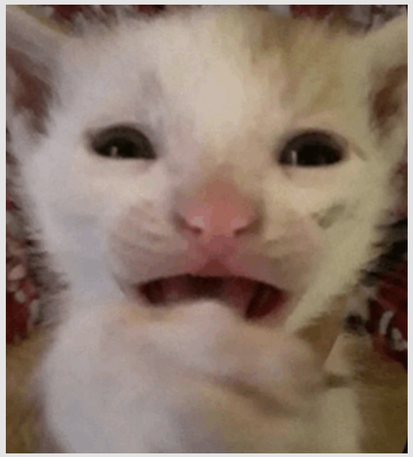
Risk Band Interpretation

- Low risk → Similar patients had ~4% mortality in test set
- Medium risk → ~17% mortality
- High risk → ~45% mortality

Alain Areeba Siddiqui
axs6903@mavs.uta.edu

Darlene Eligado
dae9134@mavs.uta.edu





Mariah Noelle Cornelio
mnc3287@mavs.uta.edu

THANK YOU